

Shadcn/ui + Next.js + Tailwind CSS

1. Project Setup

Create a Next.js JavaScript project:

```
npx create-next-app@latest shadcn-practice
```

Choose:

```
TypeScript → No  
ESLint → Yes  
Tailwind CSS → Yes  
src/ directory → Yes  
App Router → Yes  
Import alias → Yes
```

Your project:

```
shadcn-practice/  
├── src/  
│   ├── app/  
│   │   ├── layout.js  
│   │   ├── page.js  
│   │   └── globals.css  
│   └── components/  
│       └── ui/  
├── public/  
├── package.json  
└── components.json
```

Initialize shadcn:

```
npx shadcn@latest init
```

2. Button

Install:

```
npx shadcn@latest add button
```

Import:

```
import { Button } from "@/components/ui/button"
```

Basic

```
<Button>Save</Button>
```

Common Variants

```
<Button>Default</Button>
```

```
<Button variant="secondary">  
  Secondary  
</Button>
```

```
<Button variant="outline">  
  Outline  
</Button>
```

```
<Button variant="ghost">  
  Ghost  
</Button>
```

```
<Button variant="destructive">  
  Delete  
</Button>
```

```
<Button variant="link">
  Learn More
</Button>
```

Common Sizes

```
<Button size="sm">
  Small
</Button>

<Button size="default">
  Default
</Button>

<Button size="lg">
  Large
</Button>

<Button size="icon">
  +
</Button>
```

Combining variant + size

```
<Button
  variant="destructive"
  size="sm"
>
  Delete
</Button>
```

Tailwind customization

```
<Button className="rounded-full">
  Save
</Button>
```

Real-world

```
<div className="flex gap-2">
  <Button>Save</Button>

  <Button variant="outline">
    Cancel
  </Button>

  <Button variant="destructive">
    Delete
  </Button>
</div>
```

3. Card

Install:

```
npx shadcn@latest add card
```

Import:

```
import {
  Card,
  CardHeader,
  CardTitle,
  CardDescription,
  CardContent,
```

```
CardFooter,  
} from "@components/ui/card"
```

Structure

```
<Card>  
  <CardHeader>  
    <CardTitle>Profile</CardTitle>  
    <CardDescription>  
      Your profile information  
    </CardDescription>  
  </CardHeader>  
  
  <CardContent>  
    Content goes here  
  </CardContent>  
  
  <CardFooter>  
    <Button>Save</Button>  
  </CardFooter>  
</Card>
```

Important parts

```
Card  
├── CardHeader  
│   ├── CardTitle  
│   └── CardDescription  
├── CardContent  
└── CardFooter
```

Practice

Create:

Total Users
1,250
+12% this month

4. Input

Install:

```
npx shadcn@latest add input
```

Import:

```
import { Input } from "@components/ui/input"
```

Common types

```
<Input  
  type="text"  
  placeholder="Enter name"  
/>  
  
<Input  
  type="email"  
  placeholder="Enter email"  
/>  
  
<Input  
  type="password"  
  placeholder="Enter password"  
/>
```

```
<Input
  type="number"
  placeholder="Enter age"
/>

<Input
  type="date"
/>

<Input
  type="file"
/>
```

Disabled

```
<Input disabled />
```

Read only

```
<Input
  readOnly
  value="John Doe"
/>
```

5. Label

```
npx shadcn@latest add label
```

```
import { Label } from "@components/ui/label"
```

Use with Input:

```

<div className="space-y-2">
  <Label htmlFor="email">
    Email
  </Label>

  <Input
    id="email"
    type="email"
    placeholder="Enter email"
  />
</div>

```

Important:

```
htmlFor="email"
```

matches:

```
id="email"
```

6. Badge

Install:

```
npx shadcn@latest add badge
```

```
import { Badge } from "@components/ui/badge"
```

Common Variants

```

<Badge>
  Default
</Badge>

```

```
<Badge variant="secondary">
  Secondary
</Badge>

<Badge variant="outline">
  Outline
</Badge>

<Badge variant="destructive">
  Failed
</Badge>

<Badge variant="ghost">
  Draft
</Badge>
```

Real-world task status

```
<Badge>Completed</Badge>

<Badge variant="secondary">
  Pending
</Badge>

<Badge variant="destructive">
  Failed
</Badge>
```

7. Avatar

Install:

```
npx shadcn@latest add avatar
```

```
import {  
  Avatar,  
  AvatarFallback,  
  AvatarImage,  
} from "@components/ui/avatar"
```

Basic

```
<Avatar>  
  <AvatarImage src="/user.jpg" />  
  <AvatarFallback>  
    SA  
  </AvatarFallback>  
</Avatar>
```

Concept:

```
Image available  
  ↓  
AvatarImage  
  
Image unavailable  
  ↓  
AvatarFallback
```

8. Dropdown Menu

Install:

```
npx shadcn@latest add dropdown-menu
```

```
import {
  DropdownMenu,
  DropdownMenuContent,
  DropdownMenuItem,
  DropdownMenuTrigger,
} from "@components/ui/dropdown-menu"
```

Example:

```
<DropdownMenu>
  <DropdownMenuTrigger asChild>
    <Button variant="outline">
      Account
    </Button>
  </DropdownMenuTrigger>

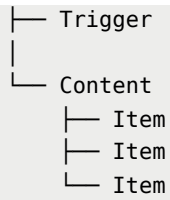
  <DropdownMenuContent>
    <DropdownMenuItem>
      Profile
    </DropdownMenuItem>

    <DropdownMenuItem>
      Settings
    </DropdownMenuItem>

    <DropdownMenuItem>
      Logout
    </DropdownMenuItem>
  </DropdownMenuContent>
</DropdownMenu>
```

Structure

```
DropdownMenu
|
```



9. Dialog

Install:

```
npx shadcn@latest add dialog
```

Import:

```
import {
  Dialog,
  DialogContent,
  DialogDescription,
  DialogFooter,
  DialogHeader,
  DialogTitle,
  DialogTrigger,
} from "@components/ui/dialog"
```

Basic

```
<Dialog>
  <DialogTrigger asChild>
    <Button>
      Add Task
    </Button>
  </DialogTrigger>

  <DialogContent>
```

```

<DialogHeader>
  <DialogTitle>
    Create Task
  </DialogTitle>

  <DialogDescription>
    Add a new task.
  </DialogDescription>
</DialogHeader>

<Input placeholder="Task name" />

<DialogFooter>
  <Button>
    Create
  </Button>
</DialogFooter>
</DialogContent>
</Dialog>

```

Real-world uses

Add Task
 Edit Task
 Delete Confirmation
 User Details
 Login
 Create Project

10. Select

Install:

```
npx shadcn@latest add select
```

```
import {
  Select,
  SelectContent,
  SelectItem,
  SelectTrigger,
  SelectValue,
} from "@components/ui/select"
```

Example:

```
<Select>
  <SelectTrigger>
    <SelectValue placeholder="Select priority" />
  </SelectTrigger>

  <SelectContent>
    <SelectItem value="low">
      Low
    </SelectItem>

    <SelectItem value="medium">
      Medium
    </SelectItem>

    <SelectItem value="high">
      High
    </SelectItem>
  </SelectContent>
</Select>
```

Real-world:

```
Priority
Status
Country
Department
```

Category
Role

11. Checkbox

```
npx shadcn@latest add checkbox
```

```
import { Checkbox } from "@components/ui/checkbox"
import { Label } from "@components/ui/label"
```

```
<div className="flex items-center gap-2">
  <Checkbox id="terms" />

  <Label htmlFor="terms">
    Accept terms
  </Label>
</div>
```

Real-world:

- ☐ Remember me
- ☐ Accept terms
- ☐ Enable notifications
- ☐ Mark as important

12. Switch

```
npx shadcn@latest add switch
```

```
import { Switch } from "@components/ui/switch"
```

```
<div className="flex items-center gap-2">
  <Switch id="notifications" />

  <Label htmlFor="notifications">
    Notifications
  </Label>
</div>
```

Good for:

- Dark Mode
- Notifications
- Email alerts
- Public profile
- Account status

13. Tabs

```
npx shadcn@latest add tabs
```

```
import {
  Tabs,
  TabsContent,
  TabsList,
  TabsTrigger,
} from "@components/ui/tabs"
```

```
<Tabs defaultValue="tasks">

  <TabsList>
    <TabsTrigger value="tasks">
      Tasks
    </TabsTrigger>
  </TabsList>
</Tabs>
```

```

</TabsTrigger>

<TabsTrigger value="projects">
  Projects
</TabsTrigger>

<TabsTrigger value="team">
  Team
</TabsTrigger>
</TabsList>

<TabsContent value="tasks">
  Tasks content
</TabsContent>

<TabsContent value="projects">
  Projects content
</TabsContent>

<TabsContent value="team">
  Team content
</TabsContent>

</Tabs>

```

14. Textarea

```
npx shadcn@latest add textarea
```

```
import { Textarea } from "@/components/ui/textarea"
```

```
<Textarea  
  placeholder="Write task description..."  
>
```

Useful for:

Description
Comments
Feedback
Messages
Notes

15. Alert

```
npx shadcn@latest add alert
```

```
import {  
  Alert,  
  AlertDescription,  
  AlertTitle,  
} from "@components/ui/alert"
```

```
<Alert>  
  <AlertTitle>  
    Success  
  </AlertTitle>  
  
  <AlertDescription>  
    Your task was created successfully.  
  </AlertDescription>  
</Alert>
```

Useful for:

```
Success
Warning
Error
Information
```

16. Table

```
npx shadcn@latest add table
```

Basic structure:

```
<Table>
  <TableHeader>
    <TableRow>
      <TableHead>Task</TableHead>
      <TableHead>Priority</TableHead>
      <TableHead>Status</TableHead>
    </TableRow>
  </TableHeader>

  <TableBody>
    <TableRow>
      <TableCell>
        Design Homepage
      </TableCell>

      <TableCell>
        High
      </TableCell>

      <TableCell>
        <Badge>Completed</Badge>
      </TableCell>
    </TableRow>
  </TableBody>
</Table>
```

```

    </TableCell>
  </TableRow>
</TableBody>
</Table>

```

17. Sheet

This is especially useful for **mobile navigation**.

```
npx shadcn@latest add sheet
```

Concept:

```

Desktop

Sidebar | Content

Mobile

≡
↓
Sheet opens
↓
Sidebar

```

Example:

```

<Sheet>
  <SheetTrigger asChild>
    <Button variant="outline">
      Menu
    </Button>
  </SheetTrigger>

  <SheetContent side="left">
    <h2>Navigation</h2>
  </SheetContent>
</Sheet>

```

```
<nav>
  <p>Dashboard</p>
  <p>Tasks</p>
  <p>Projects</p>
  <p>Settings</p>
</nav>
</SheetContent>
</Sheet>
```

18. Calendar

```
npx shadcn@latest add calendar
```

Useful for:

```
Task deadline
Appointment
Event
Project schedule
Date filtering
```

19. Common Tailwind Classes

Students should memorize these before building the final project.

Spacing

```
p-4
px-4
py-2
m-4
mt-4
mb-4
```

```
gap-4  
space-y-4
```

Flex

```
flex  
items-center  
justify-center  
justify-between  
flex-col  
gap-4
```

Grid

```
grid  
grid-cols-1  
grid-cols-2  
grid-cols-3  
grid-cols-4
```

Responsive

```
sm:  
md:  
lg:  
xl:
```

Example:

```
<div className="  
  grid  
  grid-cols-1  
  md:grid-cols-2  
  lg:grid-cols-4  
  gap-4  
">
```

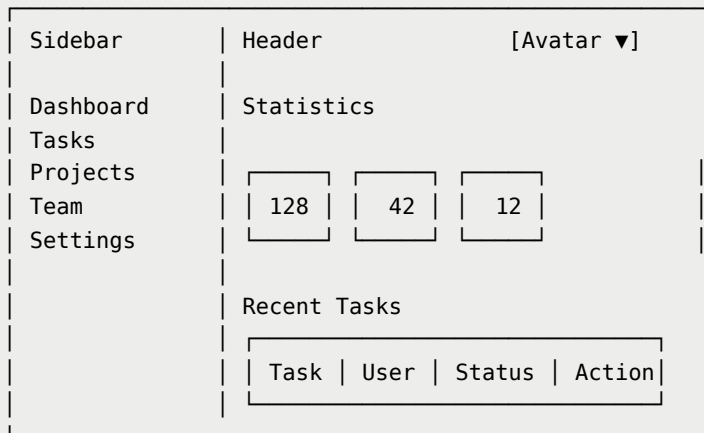
Means:

```
Mobile → 1 column  
Tablet → 2 columns  
Desktop → 4 columns
```

Final Practice Project — Task Management Dashboard

Now combine the components.

Dashboard



Components used

```
Sidebar  
Sheet  
Card  
Button  
Avatar  
Dropdown Menu  
Badge  
Table  
Dialog  
Input
```

Textarea

Select

Tabs

Switch

Checkbox
